CTA Bus Reliability Analysis

Which Chicago bus routes are least reliable, when, and what goes with it
Data: CTA bus arrival estimates (Mansueto Institute StopWatch), April to June 2026
Scope: 12 high-frequency routes, 24 route patterns, 91 days, 185,851 headways
Prepared: November 2026

1. Executive summary

A city transit authority has received complaints that some bus routes are consistently late and unpredictable, and wants answers backed by statistics rather than impressions. This report examines 12 high-frequency CTA bus routes over 91 days (April to June 2026). Because the data contains estimated arrival times but no published schedule, reliability is measured by how evenly buses are spaced along a route (the gap between consecutive buses, called the headway), which is also what riders on frequent routes actually experience.

Headline: Reliability differs strongly by route, and the biggest problem is the weekday afternoon peak (3 to 7 pm). Route 66 is the least regular route, with the most bunching and the most long gaps. Routes 3, 151, 22 and 8 deteriorate sharply in the afternoon peak, while Routes 60 and 95 stay steady all day.

Key findings

- **Routes differ a lot, and consistently.** Route identity explains about a third of the variation in daily regularity (partial eta-squared 0.32), and 60 of 66 route pairs are statistically distinguishable. Route 66 is the least regular and Routes 95 and 60 the most regular.
- **The afternoon peak is the weak spot.** Headway variability is about 0.25 higher at the PM peak than in the morning (95% CI 0.23 to 0.28), and the odds of a gap more than twice the usual are 2.8 times higher. The effect is concentrated on Routes 66, 3, 151, 22 and 8.
- **Frequency does not guarantee predictability.** Routes with the same typical frequency can differ sharply: at about 9.5 minutes between buses, Route 3 imposes about 70% more excess waiting time than Route 60. Route 66 has the most bunching (14.9% of headways under a quarter of the usual) and the most big gaps (13.9%).
- **Weekends are not worse than weekdays.** The one clear difference runs the other way: weekday mornings are less regular than weekend mornings on all 12 routes. Midday, afternoon and evening show no consistent weekday and weekend difference.
- **Slower days are less regular, but weather is not the story.** Using bus speed as a congestion proxy, each extra mph goes with lower variability (about -0.09). Rain has no detectable effect. In practice, time of day matters much more than day-to-day speed.

- **Most individual bad headways cannot be predicted.** Route and time of day are consistent drivers, but a model using all available factors explains only about 4% of the variation (McFadden R-squared 0.038), so the evidence says where to look rather than which trip will fail.

Answers to the seven questions

Question	Short answer
1. Are some routes significantly less reliable?	Yes. Large, consistent differences (partial eta-squared 0.32); Route 66 stands out.
2. Which routes have the highest average delays?	Not measurable: the data has no scheduled times. The closest proxy, excess wait, is highest on Routes 22 (3.4 min) and 151 (3.1 min).
3. Does reliability vary by time of day?	Yes. The weekday PM peak is clearly worst (+0.25 vs the AM peak).
4. Do weekdays and weekends differ?	Mostly no. Only weekday mornings are less regular than weekend mornings.
5. Does congestion contribute?	Slower-than-usual days are less regular (association, using speed as a proxy). Rain shows no effect.
6. Are some routes unpredictable even if frequent?	Yes. Routes 60, 4, 49 and 3 all run about every 9.5 minutes but differ in spread (SD 5.2 to 6.7 min). Route 66 has the most bunching and gaps.
7. What would improve reliability?	See Section 4: PM-peak spacing control, targeted work on Routes 66, 3, 151, 22 and 8, and a piloted evaluation.

2. Data and approach

2.1 Data source

The analysis uses the Mansueto Institute's StopWatch dataset for the Chicago Transit Authority. CTA does not publish a record of when each bus actually reached each stop. StopWatch reconstructs estimated stop-level arrival times by interpolating between bus locations scraped from the CTA Bus Tracker every five minutes. Each route pattern (one direction or variant of a route) has its own file.

The dataset has no scheduled arrival times, so the brief's "delay minutes" and "late arrival" could not be computed directly. The table below shows what was used in their place.

In the brief	Used in this analysis
Delay = actual minus scheduled arrival	Headway: minutes since the previous bus at the same stop, divided by that pattern's usual headway for the same day type and hour ("normalized")
Late arrival (more than 5 minutes)	Big gap: headway more than 2x the usual. Bunched: headway under 25% of the usual. A fixed 5-minute version is also reported.
Traffic level (low, medium, high)	Bus speed relative to the route's own usual speed (a coarse congestion proxy)
Weather	Daily rainfall and mean temperature for Chicago (Open-Meteo)
Passenger count, cancellations	Not available at trip level

2.2 Route selection and exclusions

Twenty well-known frequent routes were shortlisted and the two busiest active patterns of each were examined. Routes 53 and 63 were dropped for too little service (about 8 and 16 trips a day). Routes 79, 9, 77, 12, 20 and 55 were dropped because buses were missing from the data in many hours (for example, Route 79 showed about one bus an hour at midday against about six on a healthy route), which would have made them look falsely unreliable. The final set is 12 routes: 3, 4, 8, 22, 36, 49, 50, 60, 66, 74, 95 and 151.

2.3 Statistical approach

- **Independence.** A bus passes dozens of stops, so using every stop would hugely overstate the sample. One mid-route reference stop per pattern is used, and the two directions are averaged into one value per route, day and period.
- **Honest uncertainty.** Routes share each day's conditions, so standard errors are clustered by date and bootstrap intervals resample whole days. With about 91 days, intervals are somewhat optimistic.
- **Tests.** Welch's ANOVA and Kruskal-Wallis (unequal variances, skewed data), regression with route fixed effects, Friedman tests for repeated measures, and logistic regression for big gaps. Multiple pairwise comparisons use Holm correction.
- **Effect sizes.** Partial eta-squared, Kendall's W, Cohen's d and odds ratios are reported alongside p-values, because with thousands of observations even tiny differences are "significant".

3. Findings

3.1 Routes differ in regularity (Investigations 1 and 2)

Figure 1 ranks routes by normalized headway variability, where higher means buses are spaced less evenly relative to the schedule. Intervals come from resampling whole days.

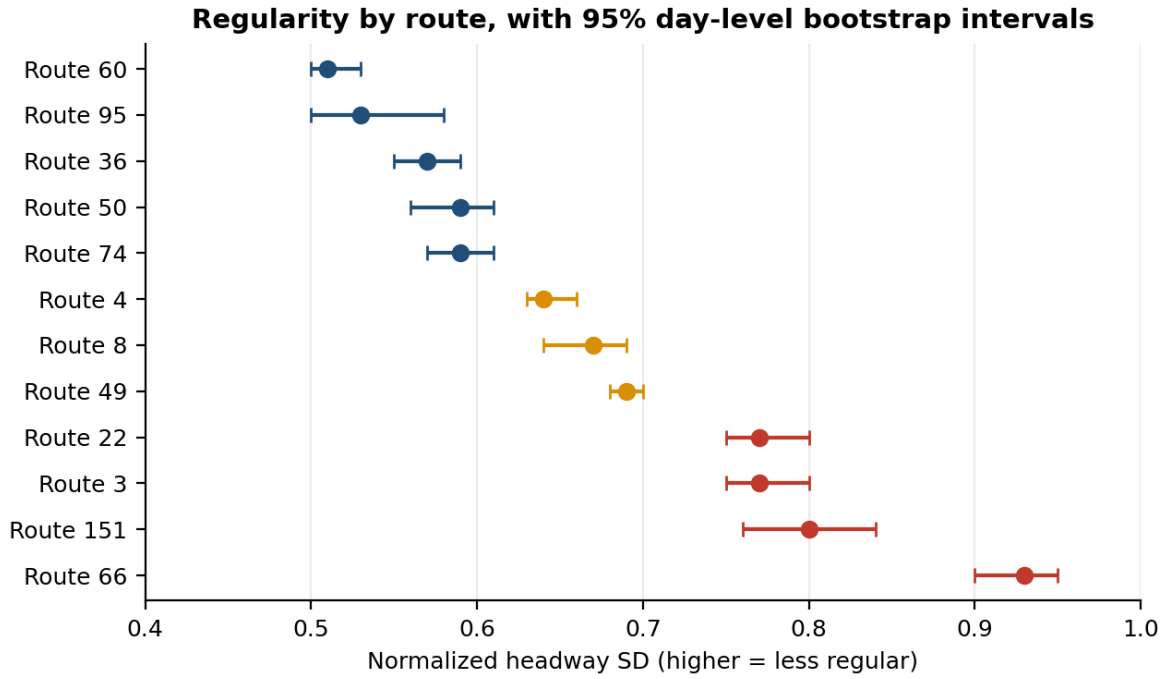


Figure 1. Normalized headway standard deviation by route (April to June 2026).

Routes fall into bands. Routes 60 (0.51) and 95 (0.53) are the most regular. Routes 36, 50 and 74 (0.57 to 0.59) are good. Routes 4, 8 and 49 (0.64 to 0.69) are middling. Routes 22, 3 and 151 (0.77 to 0.80) are poor, and Route 66 (0.93) is clearly the worst. Adjacent routes within a band overlap, so they should not be ranked against each other.

With each date and time period held fixed, route explains about a third of the remaining variation in regularity (F = 164, partial eta-squared = 0.32). A rank-based test agrees (Friedman chi-square = 1,214, Kendall's W = 0.34), and 60 of 66 route pairs differ after correction. The pairs that cannot be told apart are 151 and 22, 3 and 49, 22 and 49, 50 and 74, 36 and 60, and 4 and 8.

Robustness. The same routes sat at the top and bottom whether the analysis used one month or three, and whether the questionable routes were excluded or not.

3.2 Predictability, not just averages (Investigation 6)

Two routes can have the same typical frequency and very different reliability. Figure 2 plots each route's median headway against the spread of its headways. Routes 60, 4, 49 and 3 all run about every 9.5 minutes, yet their standard deviations rise from 5.2 to 6.7 minutes. Route 3 also imposes about 70% more excess waiting time (2.2 minutes against 1.3) on a rider who arrives at random.

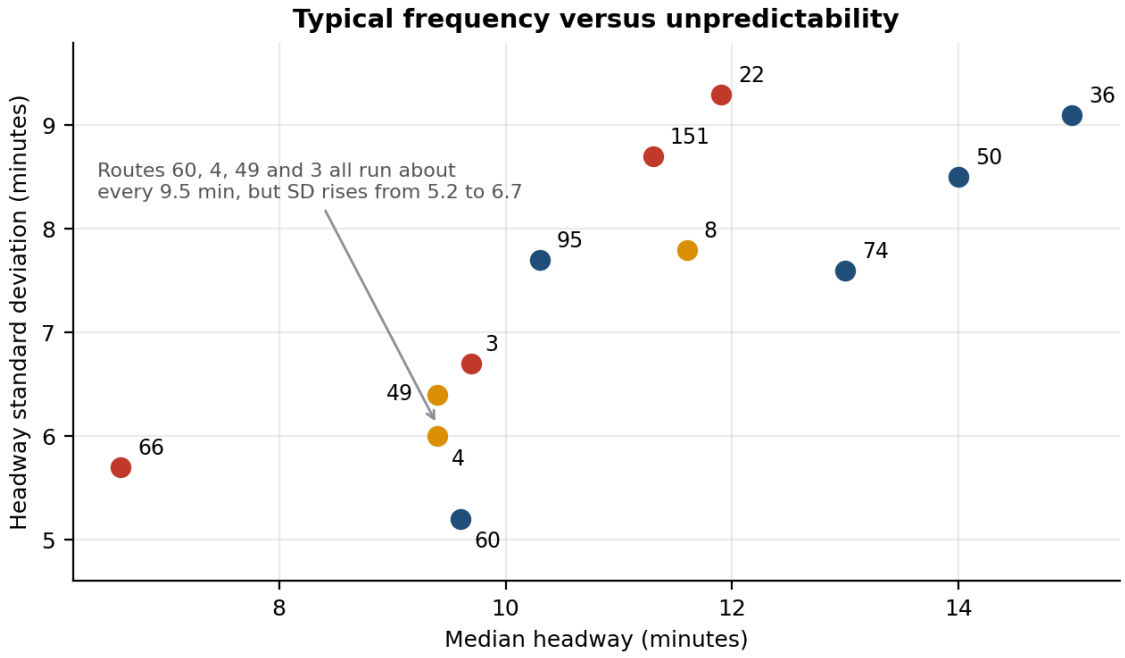


Figure 2. Median headway against headway standard deviation, one point per route.

Figure 3 shows how often buses are bunched (arriving in under a quarter of the usual headway) or leave a big gap (over twice the usual). Route 66 is highest on both (14.9% and 13.9%): buses travel in pairs and leave long holes behind them. Routes 151, 22 and 3 are also elevated on both measures (12.5% to 13.9% bunched, 9.1% to 11.7% big gaps), roughly double Route 60 (6.1% and 4.1%). Route 95 has the lowest bunching (4.5%) but a higher gap rate (6.4%).

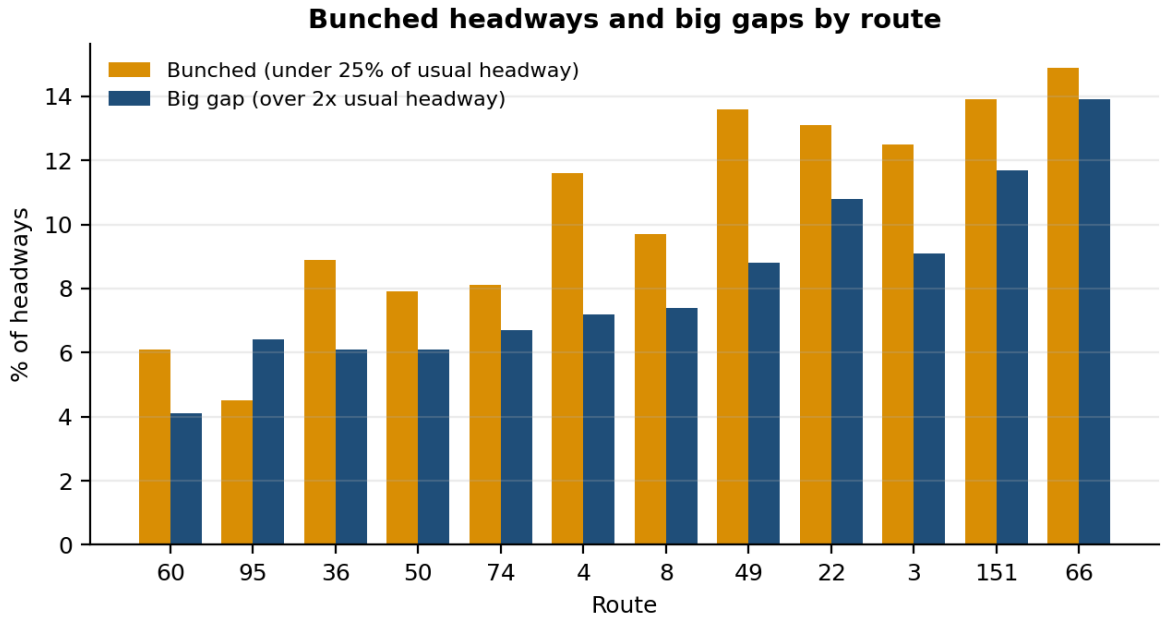


Figure 3. Share of headways that are bunched or big gaps, by route.

3.3 Time of day (Investigation 3)

On weekdays, the PM peak (3 to 7 pm) is the least regular period. Compared with the AM peak, headway variability is higher by 0.254 at the PM peak (95% CI 0.232 to 0.275), 0.083 at midday and 0.079 in the evening (Figure 4). The result was the same when headways were normalized by hour instead of by period, which rules out schedule ramp-up as the cause.

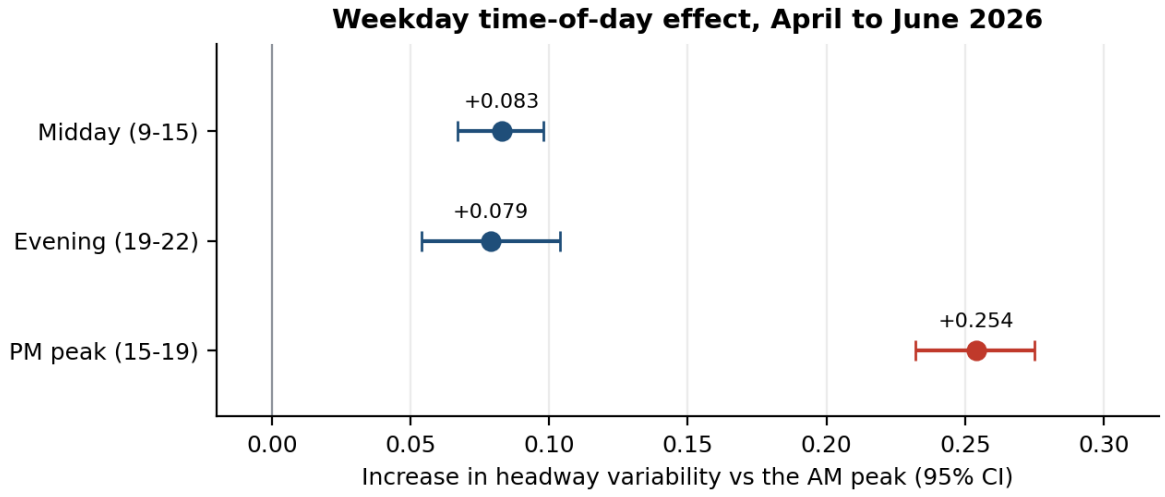


Figure 4. Weekday change in headway variability relative to the AM peak, with 95% confidence intervals.

The size of the afternoon problem differs a great deal by route (Figure 5, which shows the share of headways more than twice the usual). Route 3 goes from 3.4% in the morning to 18.3% in the PM peak. Route 66 reaches 22.5% in the PM peak and is poor in every period. Routes 151, 22 and 8 follow the same pattern more mildly. Route 60 stays under 6% all day. Route 95 is the exception, with its worst period in the morning (8.7%), which suggests a different cause on that route.

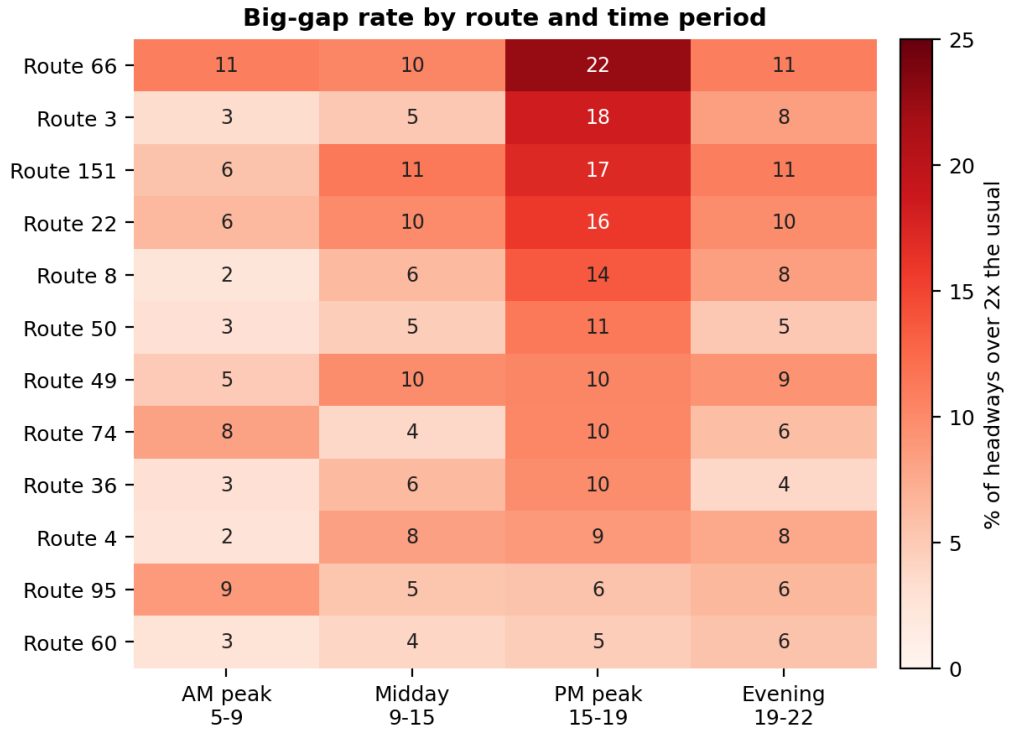


Figure 5. Big-gap rate (% of headways over twice the usual) by route and time period.

A route by period interaction test confirms that the afternoon penalty is not uniform across routes (F = 16.3, p < 0.0001, on the April data).

3.4 Weekdays and weekends (Investigation 4)

The brief asked whether service becomes less reliable on weekends. In this data it does not. Overall, weekends are slightly more regular (difference -0.035, 95% CI -0.061 to -0.009), but Figure 6 shows that this comes almost entirely from the morning. Weekend mornings are more regular than weekday mornings on all 12 routes (difference 0.176, Wilcoxon p = 0.0005). In the other three periods there is no consistent difference.

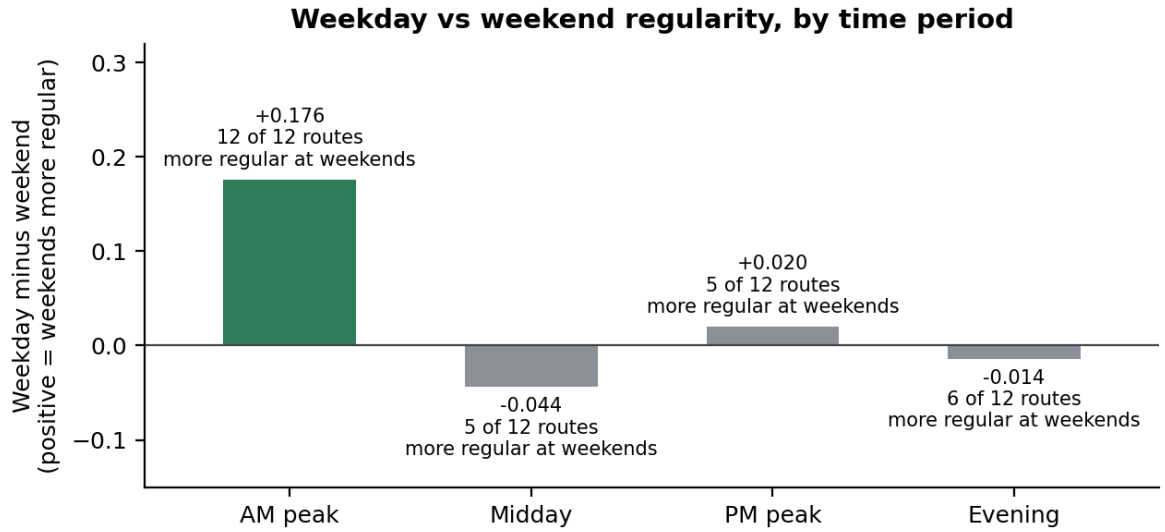


Figure 6. Weekday minus weekend regularity by period (positive = weekends more regular).

A correction worth noting. In April alone the overall weekend advantage was about 0.08, more than double the three-month estimate of 0.035, so the broader weekend result shrank as data was added. The narrow weekday-morning result is the one to rely on. Only 26 weekend days are in the sample.

3.5 Congestion proxy and weather (Investigation 5)

There is no direct traffic measure, so bus speed (relative to a route's own usual speed) stands in for congestion. Days when buses move more slowly than usual are less regular: each extra mph goes with a 0.091 lower headway variability (95% CI 0.069 to 0.113 lower), after adjusting for route, time period, weekend and weather. Figure 7 shows the same thing in simpler terms for the first month of data.

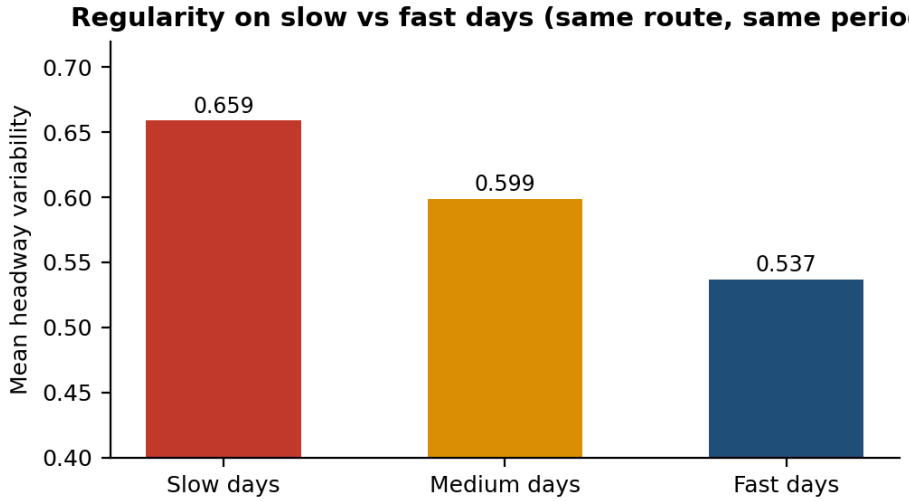


Figure 7. Mean headway variability on slow, medium and fast days (April 2026, within route and period).

Rain made no detectable difference: on days with at least 5 mm of rain (20 of 91 days) the estimated effect on regularity was -0.003 (95% CI -0.029 to +0.022).

How to read this. A typical day-to-day speed swing is only about 0.5 mph, which moves variability by about 0.05, against 0.25 for the PM-peak effect. Speed is also not a pure traffic measure (it reflects stop spacing, signals and boarding) and the link may run both ways, since bunched buses slow each other down. It is an association, not proof that traffic causes irregularity.

3.6 What predicts a big gap? (Investigation 7)

A logistic regression models the chance that a headway is more than twice the usual, using route, time period, weekend, speed and weather, with date-clustered standard errors. Relative to Route 60, the odds are 3.7 times higher on Route 66, 3.1 times on Route 151 and 2.8 times on Route 22 (Figure 8).

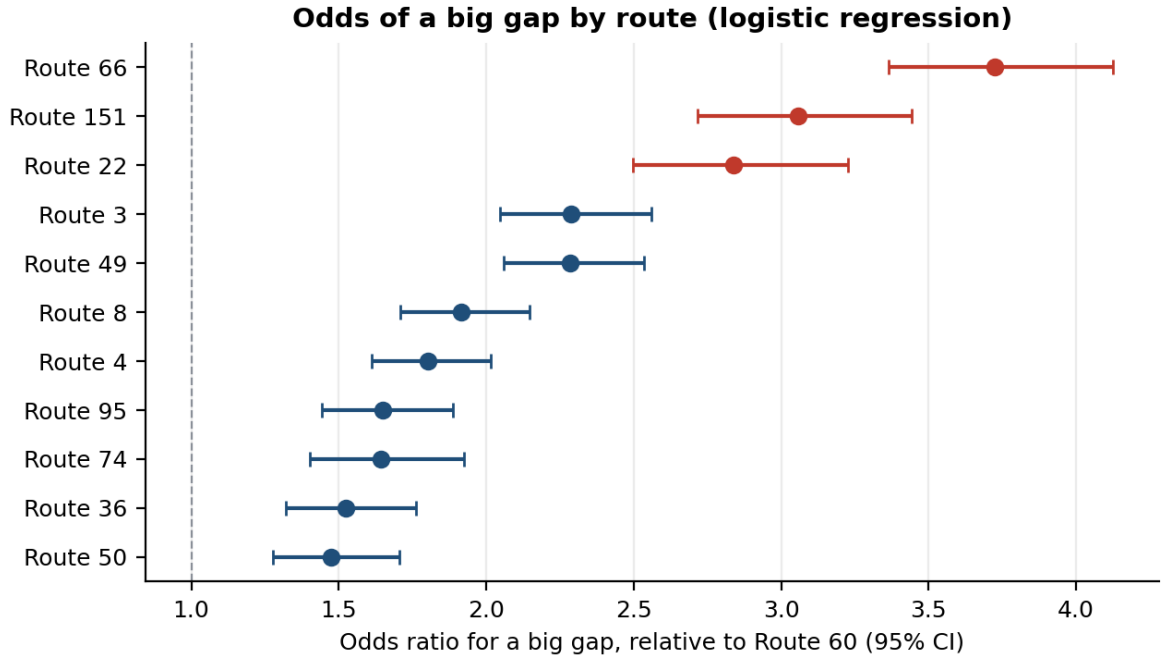


Figure 8. Odds of a big gap by route relative to Route 60 (logistic regression, 95% CI).

Factor	Odds ratio	95% CI	Reading
PM peak vs AM peak	2.81	2.56 to 3.07	Strongest time-of-day effect
Evening vs AM peak	1.63	1.47 to 1.82	Higher odds
Midday vs AM peak	1.51	1.37 to 1.66	Higher odds
Speed, +1 mph above usual	0.69	0.57 to 0.85	About 31% lower odds
Weekend	1.20	1.03 to 1.39	Small; no consistent pattern in the other analyses
Rain (5 mm or more)	0.96	0.88 to 1.04	No clear effect

Two cautions. First, the model explains very little (McFadden R-squared 0.038), so most individual bad headways are not predictable from these variables. Second, a version using a fixed definition (more than 5 minutes above the usual headway) ranks routes differently: Routes 22, 151 and 36 look worst and Route 66 looks nearly fine, because five minutes is a far bigger deviation on a 6.6-minute headway than on a 15-minute one. The relative definition is the fairer way to compare routes with different frequencies.

4. Recommendations

These follow from the evidence above. All are based on observational data, so they identify where service is weakest, not proven causes.

1. **Target the weekday PM peak.** It is the least regular period on 11 of 12 routes. Apply spacing-based control in the afternoon: dispatch and hold buses by headway rather than strictly by timetable, and put field supervisors at key points on Routes 66, 3, 151, 22 and 8.
2. **Prioritize Route 66.** It has the highest bunching and gap rates and is poor in every period, not just the afternoon. Candidate measures are holding at timepoints to restore spacing, transit signal priority at bottleneck intersections, and a review of boarding and dwell patterns.
3. **Treat Routes 3, 151, 22 and 8 as PM-peak problems.** Their big-gap rate rises sharply in the afternoon (for example Route 3 from 3.4% to 18.3%), so look at spare vehicles and recovery time at layovers, missed trips, and operator availability in the afternoon.
4. **Use Routes 60 and 95 as benchmarks.** They stay steady all day. Compare how they are scheduled, dispatched and supervised with Routes 66 and 3, and which of those practices transfer.
5. **Investigate the weekday morning.** Every route is more regular on weekend mornings. Examine whether lighter traffic, simpler schedules or different dispatch practices explain it, since that could point to a fix for weekday mornings.
6. **Pilot, then evaluate.** Choose an intervention route and a comparable control route, measure the same headway metrics before and after, and use that comparison to judge whether a change works.
7. **Fix the data first where needed.** Routes 79, 9, 77, 12, 20 and 55 could not be analyzed because buses were missing from the data. Obtain official schedule-adherence data, hourly weather, traffic speeds and passenger counts so that delay, congestion and demand can be analyzed directly.

5. Limitations

- **Arrival times are estimates.** They are interpolated from five-minute location pings, so small timing differences are unreliable and there is no true schedule delay. Headway regularity is a valid reliability measure for frequent service but is not the same as punctuality.
- **Only 12 routes.** Eight of the 20 shortlisted routes were excluded. If missing data coincides with real service problems, exclusions may flatter the network. The routes analyzed are frequent main-line routes, not a sample of all CTA routes, and fewer than the 30 in the original brief.
- **Short period.** Three months, with about 91 effectively independent days, because all routes share each day's conditions. Weekend results rest on only 26 days.
- **Thresholds are conventions.** Big gap (over 2x usual) and bunched (under 25% of usual) are reasonable but not unique choices. Gaps over 60 minutes and repeated pings from the same vehicle were removed.
- **Proxies.** Speed is a segment speed shared across several stops and mixes traffic with stop spacing and boarding. Weather is a daily total, so rain during rush hour is not captured.
- **Observational data.** Associations are not proof of cause, and many tests were run. Pairwise results use Holm correction, but individual numbers should be read as estimates.

6. Conclusion

Bus reliability on the analyzed CTA routes is neither uniform nor random. Differences between routes are large and persistent, differences by time of day are clear, and both are far stronger than anything attributable to weekday versus weekend or to rain. The weekday afternoon peak is where service spacing breaks down most, and it does so unevenly: Route 66 is poor throughout the day, Routes 3, 151, 22 and 8 deteriorate sharply in the afternoon, and Routes 60 and 95 show that steady service is achievable on this network.

The findings support focusing operational effort where the evidence is strongest: afternoon spacing control on the five worst routes, route-specific treatments, and a piloted evaluation. They also show the limits of what the available data can say. With better schedule, traffic and demand data, the same analysis could quantify delay directly and test why the afternoon peak fails.

Appendix A. Route summary

n = headways analyzed. One mid-route reference stop per pattern; intervals are 95% day-level bootstrap intervals; sorted from most to least regular.

Route	n	Median (min)	SD (min)	Excess wait (min)	Normalized SD (95% CI)	Bunched %	Big gap %
60	18,103	9.6	5.2	1.3	0.51 (0.50 to 0.53)	6.1	4.1
95	10,112	10.3	7.7	2.5	0.53 (0.50 to 0.58)	4.5	6.4
36	11,636	15.0	9.1	2.7	0.57 (0.55 to 0.59)	8.9	6.1
50	11,810	14.0	8.5	2.5	0.59 (0.56 to 0.61)	7.9	6.1
74	13,538	13.0	7.6	2.2	0.59 (0.57 to 0.61)	8.1	6.7
4	17,995	9.4	6.0	1.8	0.64 (0.63 to 0.66)	11.6	7.2
8	14,237	11.6	7.8	2.5	0.67 (0.64 to 0.69)	9.7	7.4
49	18,959	9.4	6.4	2.1	0.69 (0.68 to 0.70)	13.6	8.8
22	13,296	11.9	9.3	3.4	0.77 (0.75 to 0.80)	13.1	10.8
3	16,986	9.7	6.7	2.2	0.77 (0.75 to 0.80)	12.5	9.1
151	14,606	11.3	8.7	3.1	0.80 (0.76 to 0.84)	13.9	11.7
66	24,573	6.6	5.7	2.2	0.93 (0.90 to 0.95)	14.9	13.9

Appendix B. Glossary

- **Headway:** minutes between consecutive buses at the same stop.
- **Normalized headway SD:** the standard deviation of headway divided by the usual headway for that route, day type and hour. It allows routes with different frequencies to be compared; 0 would be perfectly even spacing.
- **Bunched:** a headway under 25% of the usual, meaning buses arriving almost together.
- **Big gap:** a headway more than twice the usual.
- **Excess wait:** the extra time, in minutes, a rider who arrives at random waits because buses are unevenly spaced (variance of headway divided by twice the mean headway).
- **Partial eta-squared and Kendall's W:** effect sizes, from 0 (no effect) towards 1 (strong effect). Partial eta-squared of 0.32 is a large effect.
- **Odds ratio:** how many times higher the odds of an event are compared with a baseline; 1 means no difference.
- **Clustered standard errors:** uncertainty estimates that allow observations from the same day to be correlated, which prevents overstating confidence.
- **Holm correction:** an adjustment to p-values when many comparisons are made, to avoid false positives.

Note on figures: charts are drawn from the summary tables produced by the analysis notebook, not re-estimated from raw data. Figure 7 uses April 2026 only; the others use April to June.